

Question No:2

DS	Session	Sorting	Question Information	Level 1	Challenge 12
Question description Sajid is an third year student in a reputed institution. Although he scored well in many subjects, he did not an expert in computer programming languages. But Sajid's computer examination is scheduled for next week. As per the blueprint, many questions would come from the sorting topic. He collected previous year's questions. one of the repeated questions is to sort the given set of numbers using Selection Sort The first line of the input contains the number of elements N, the second line of the input contains the numbers A_i to be sorted. In the output print the the final sorted array in the given format. Can you help him ? Constraints $1 \leq N \leq 10^5$ $1 \leq A_i \leq 10^9$ Input: The first line of the input contains the number of elements the second line of the input contains the numbers to be sorted. Output: print the the final sorted array in the given format.					

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

5
15 17 11 25 1

EXPECTED OUTPUT

1 11 15 17 25

Test Case 2

INPUT (STDIN)

6
65 45 27 21 25 11

EXPECTED OUTPUT

11 21 25 27 45 65

▼ Mandatory Test Cases

Test Case 1

KEYWORD

void selectionSort(int
arr[],int n)

Test Case 2

KEYWORD

void swap(int *xp,int *yp)

Test Case 3

KEYWORD

void printArray(int arr[],int
size)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

275

Test Case 3

NLOC

47

Code:

```
1  #include <stdio.h>
2  int main()
3  {
4      int n;
5      scanf("%d", &n);
6      int a[n];
7      // Read the elements
8      for(int i = 0; i < n; i++)
9      {
10         scanf("%d", &a[i]);
11     }
12     // Selection Sort
13     for(int i = 0; i < n - 1; i++)
14     {
15         int min = i;
16
17         for(int j = i + 1; j < n; j++)
18         {
19             if(a[j] < a[min])
20             {
21                 min = j;
22             }
23         }
24         // Swap
25         int temp = a[i];
26         a[i] = a[min];
27         a[min] = temp;
28     }
```

```
29     // Print sorted array
30     for(int i = 0; i < n; i++)
31     {
32         printf("%d ", a[i]);
33     }
34     return 0;
35 }
```

Output:

Test Case-1:

```
5
15 17 11 25 1
1 11 15 17 25
```

Test Case-2:

```
6
65 45 27 21 25 11
11 21 25 27 45 65
```

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